

README

Helix Cluster OS

Helix Cluster OS is a next-generation distributed operating system for orchestrating compute workloads across heterogeneous nodes — from datacenter GPUs down to edge SBCs and handhelds. It unifies HPC scheduling, container orchestration, AI/ML inference, federated multi-cluster operation, and secure multi-tenant session management under a single control plane.

Engineering guarantee (CLAUDE-1 / CLAUDE-2): every feature ships with tests that prove real end-user behaviour — never green tests over stubs — and every OS-specific capability uses a real native facility per platform (no Linux-only mocks). See [CLAUDE.md](#) and [Constitution.md](#).

Features

- **Heterogeneous Node Management** — Register, monitor, and schedule across tiers T1–T8 (datacenter → microcontroller) with structured health scoring and capability negotiation.
- **Omega-model Scheduler** — Pluggable placement with optimistic concurrency, ClassAd matching, gang scheduling, value-multiplier preemption, multifactor (age/fairshare/size/QoS) priority queues, and constraint-based (location/colocation/order/stickiness) placement.
- **Distributed-Systems Foundation** — A large pure-Go library (pkg/) of consensus, membership (SWIM gossip), replication (CRDT, MVCC, anti-entropy), and federation primitives. See [Foundation Packages](#).
- **Federation & Multi-cluster** — Cross-cell trust, topology patterns, CRDT config sync, data-residency admission, split-brain detection, and quorum-based failure confirmation.
- **GPU & Cost Orchestration** — GPU pool management, TCO-aware local cost modelling, cost/latency-aware schedulers, burst-to-cloud autoscaling, N+K failover capacity reserve, and global budget caps.
- **Security & Attestation** — SPIFFE identity, JWT auth, ML-KEM-768 post-quantum E2EE, device attestation (challenge/response, proof-of-GPU-work, device sealing), and attestation-gated admission.
- **Deterministic Simulation Testing (DST)** — FoundationDB-style seeded simulation, BUGGIFY fault injection, Turmoil network simulation, and clock-fault injectors for reproducible distributed-systems testing.
- **Observability** — Built-in metrics, W3C distributed tracing, structured logging, and Grafana dashboard generation.
- **Multi-Protocol APIs** — gRPC services with Protocol Buffer definitions for all subsystems.

Architecture

Helix is organised as a seven-layer stack (L0–L7) with 14 control-plane microservices, coordinated via SWIM gossip for membership and Raft consensus for strongly-consistent state. The scheduler is an Omega-model two-level design with optimistic concurrency.

The repository is a Go workspace combining a core module with git submodules for the larger services.

```
.
├─ api/v1/           # Protocol Buffer definitions (NodeService,
SessionService, SchedulerService, ...)
├─ cmd/              # Service binaries and CLIs
├─ internal/         # Private application packages (console,
gateway, scheduler, node, health, policy, trust, ...)
├─ pkg/              # Shared pure-Go foundation library (see
"Foundation Packages")
├─ web/              # React + TypeScript + Vite dashboard
├─ docs/             # Documentation (see "Documentation")
├─ data/             # HXC registry (SQLite) and runtime data
├─ scripts/          # Build, test, and utility scripts
├─ HelixConstitution/ # Governance & constitution (submodule)
├─ security/         # Security service: identity, E2EE,
attestation (submodule)
├─ helixqa/          # HelixQA challenge/validation framework
(submodule)
├─ EventBus/         # Event bus (submodule)
├─ Messaging/        # Messaging service (submodule)
├─ discovery/        # Service discovery (submodule; pkg/
discovery is the editable root)
├─ containers/       # Container runtime (submodule)
├─ recovery/         # Recovery service (submodule)
├─ config/           # Configuration service (submodule)
├─ challenges/       # Challenge platform (submodule)
├─ DocProcessor/     # Document processing (submodule)
├─ LLMOrchestrator/  # LLM orchestration (submodule)
├─ Herald/           # Notification service (submodule)
├─ docs_chain/       # Documentation-chain engine (submodule)
```

Foundation Packages

The pkg/ library provides the pure-Go, deterministic, well-tested primitives the control plane is built from. Highlights by domain:

Domain	Packages
Consensus & coordination	voting (largest-subcluster quorum), failconfirm (SWIM two-phase PFAIL→FAIL), leader, lock, splitbrain / splitbrainalert,

Domain	Packages
	heartbeatcoalescer (Multi-Raft), multiraft (per-shard etcd-raft groups + LeaseTracker leaseholder local reads, throughput scales with shard count), stonith (STONITH fencing: IPMI/EC2/Azure/SBD + multi-level fallback), kraft (KRaft-style self-managed Raft metadata quorum, no ZooKeeper)
Membership & discovery	swim (+ phi-accrual, hierarchical), discovery (+ federated, + mDNS/ DNS-SD _helix-cluster._tcp advertiser/browser: TXT cellid/nodeid/ wgpkey, reject-invalid; discovery-only, trust via SPIFFE), nataversal (STUN), ice, cellmesh, scan (Oracle-SCAN stable virtual endpoint)
Replication & state	crdt (+ merkle, LWW/ORSet/G/PN-counters/vector-clock), deltacrd (delta-state G/PN/OR-set/LWW-map), mvcc (B-tree time-travel store), antientropy (hinted-handoff + read-repair + Merkle diff), watchmanager (syncd/unsyncd/victim), hlc, offlinesync, checkpoint_merge
Scheduling & placement	scheduler (Omega/ClassAd/gang/ preempt), constraints (Pacemaker 4-type), preempt (value-multiplier), priorityqueue (multifactor aging), backfill (SLURM), admissioncontrol (N+K reserve), budgetcap, qos, suitability, ewmarank, workclaim (SKIP LOCKED), providerchain (multi-tier fallback cascade), modelrouter (strategy→default model), gepetto (local-vs-Chutes arbitration), llmfailover (error-classified failover taxonomy), carbonsched (carbon-aware placement + per-job kWh/gCO2 metering), workloadrouter (UnifiedManager concurrent-pricing weighted composite routing + TEE multiplier)
GPU & resource mgmt	pool, local (TCO), costsched, latencysched, healthmonitor, gputest (attestation crypto), capability, deviceprofile, device, devicecatalog (machine-readable device taxonomy → tier/trust/ compute-class lookup), tierdef,

Domain	Packages
Federation & multi-cluster	tiersec, quantization, gpucatalog (compute-multiplier catalog + attested scoring), gputopo (NUMA/NVLink topology-aware placement), balancemonitor (USD balance floor warning), deviceplugin (gRPC device-plugin / GRES fingerprinting + oversubscription rejection), gpu (ProviderAdapter registration hooks → pool-facing provider listing), benchmark (real repeatable on-host CPU score + GPU TFLOPS / NPU TOPS seams), tierdetect (cross-platform host-capability pre-provision gate → typed MissingCapabilityError), internal/gpu dual-workload reservation (Helix-PoW reserve + >0.80 starvation guard) federation (+ suspicion), internal/federation (Karmada PropagationPolicy/OverridePolicy engine: constraint-aware two-level cell selection + <60s failover reselect), gitops (ArgoCD ApplicationSet client: matrix per-cell generation + prune/self-heal + canary→tier-2→tier-1 rolling sync + drift/prune), fedtopology, fedtrust, configsync, residency, raftprofile, spiffefed, doublecrypt
Messaging & flow	flowcontrol (K8s APF), workqueue (rate-limited), ratelimit, backoff, retry, idempotent (exactly-once), rebalance (cooperative-sticky), fiber, fallbackchain, pubsub, events
Routing & sessions	hashslot (CRC16 + MOVED/ASK), session, slotmigration (atomic live migration), edge, edgeregistry, edgeverify, edgefusion
Security & verification	crypto, jwt, hybridkex (X25519 + ML-KEM-768 hybrid post-quantum key exchange), e2eebench (hybridkex handshake-latency benchmark, median <1ms), modelintegrity (SHA-256 gate), redundantexec (BOINC trust), attestadmit, doublecrypt, spiffefed, gravaladmit (HMAC GraVal admission), gravalverify

Domain	Packages
	(VRAM-ratio attestation + BatchVerify KPI), gpuattest (challenge/response + seal + multi-GPU node enumeration), fsresidency (per-range ReadAt + SHA-256 file-residency challenge), exportcontrol (country-tier KYC gate on controlled-GPU node onboarding), compliancedoc (EU AI Act model-card / provenance doc-gen from attestation logs)
	burst (hysteresis autoscaler), bursthysteresis (MONITOR→SPILL→RECOVER dead-band), cloudspot (AWS IMDSv2 / Azure scheduled-events / GCP preemption interruption pollers → drain/checkpoint/upload, httptest-proven), marketplace, marketplaceadapter (MarketplaceAdapter interface + Name()-dispatch registry + Chutes HTTP + Akash/AKT adapters), revenueopt (greedy GPU→marketplace revenue maximiser, TEE→Chutes bias), economics (multi-token RewardDistributor with treasury/reinvest conservation + participant ROI/break-even), chutesaccount (Chutes API model-list + balance client), chutes (inference client + attestation + E2EE envelope + ChutesMinerConfig/ValidatorConfig validation), provider/chutes (OpenAI-compatible /v1 provider + 429 retry/backoff + Retry-After), provider/runpod (serverless warm-pool GPUProvider), provider/aws (EC2 Spot GPUProvider, injectable client), provider/ionet (io.net REST Ray-cluster GPUProvider: DeployCluster/HealthCheck + capacity gate, httptest-proven), llmadapter (Claude/OpenAI request/response shape adapters)
Burst & economics	testing/dst (+ BUGGIFY, chaos, turmoil), dst (standalone seeded-RNG deterministic sim harness + byte-for-byte replay), porcupine (WGL linearizability checker + recorder), internal/chaos (PodKill/partition/disk-stall/clock-skew injectors + canary
Testing & simulation	

Domain	Packages
	rollback), timefault, chaosexp, fmea, phasegate, qualitygate, phase7matrix, stats, covgate, sandbox
Observability	metrics (+ tier/cost/provider-health series, + TAO-earnings/GraVal-status/token-throughput/gpu-utilization series), tracing (W3C), health (+ miner-api/GraVal DaemonSet dependency checks named in the rollout), grafanadash, log

Each package is standard-library-only where possible, deterministic (injected clocks, seeded PRNGs), and proven by tests that fail under mutation of the logic they cover.

Quick Start

Prerequisites

- **Go 1.26+** (workspace uses `go.work`)
- Node.js 20+ (for the web UI)
- SQLite 3 (the HXC registry lives at `data/hxc_registry.db`)
- Docker & Docker Compose (for integration services)
- Protocol Buffer compiler + `protoc-gen-go` (optional, for API regeneration)

Setup / Build / Test

```
./scripts/setup.sh      # initialise submodules and toolchains
./scripts/build.sh      # go build ./...
./scripts/test.sh       # go test ./...
./scripts/lint.sh       # go vet + linters
./scripts/format.sh     # gofmt
```

To run the full race suite for a package:

```
go test -race -count=1 ./pkg/<package>/...
```

API

Protocol Buffer definitions live in `api/v1/`. Core services:

- **NodeService** — Node lifecycle management
- **SessionService** — Session CRUD operations
- **SchedulerService** — Job scheduling and monitoring
- **HealthService** — Health checks and reporting
- **AdvisoryService** — Distributed locks and advisory events
- **SecurityService** — Authentication and authorization
- **BuildService** — Build pipeline management

Web UI

The web dashboard is built with React, TypeScript, and Vite.

```
cd web && npm install && npm run dev
```

Documentation

- [CLAUDE.md](#) — AI-agent engineering rules (end-user usability & cross-platform parity guarantees)
- [Constitution.md](#) / [HelixConstitution/](#) — project governance
- [docs/ARCHITECTURE.md](#) — hardened L0–L7 architecture diagram + component map (lint-enforced by [pkg/archlint](#))
- [docs/MVP_ARCHITECTURE.md](#) — living MVP seven-layer overview + service-communication Mermaid grounded in real packages, with a Go drift-validator that fails the build on doc/codebase drift (HXC-1145)
- [docs/FOUNDATION_PACKAGES.md](#) — full catalogue of pkg/ packages
- [docs/HXC_REGISTRY.md](#) — the work-item registry model
- [docs/NODE_PROVISIONING_BOUNDARY.md](#) — node-provisioning boundary: Helix provisions operator-controlled nodes; it does **not** jailbreak/root/unlock or bypass any device security (HXC-1146)
- [docs/guides/phase_02_architecture.md](#) — Phase 02 architecture guide: operator console → WireGuard/SWIM mesh → discovery/scheduler → Linux node, with a traced job path and the no-jailbreak boundary (HXC-1164)
- [CODING_STANDARDS_GO.md](#) / [CODING_STANDARDS_C.md](#) / [CODING_STANDARDS_ZIG.md](#) — language standards
- [DEVELOPMENT.md](#) — development workflow
- [CHANGELOG.md](#) — release history

Documentation Synchronization (CLAUDE-3 / §11.4.106)

Every change to code, services, components, architecture, or schema **MUST** update all affected materials — README, docs, user guides, manuals, websites, diagrams, and SQL/schema definitions — together with **all** of their exports. This is mechanically enforced out-of-the-box by the docs_chain engine via `.docs_chain/contexts/*.yaml` (Markdown → HTML/PDF/DOCX) and gated by docs_chain verify, with no escape hatch. The mandate restates and cites Constitution §11.4.106 and the project rules CLAUDE-3 / AGENT-2 / QWEN-2.

Development Model

Work is tracked in an SQLite registry (`data/hxc_registry.db`) of HXC-#### items across 11 phases (0–10). Items move Queued → In progress → Completed and are implemented in parallel “waves”: disjoint new packages built via an implement → adversarial-review → fix pipeline, then gated with whole-tree build/vet, -race tests, and an independent mutation bite per item that must fail the item’s named guard test. No item is marked Completed without that proof.

Contributing

1. Create a feature branch
2. Implement with tests that prove real behaviour (and fail under mutation)
3. Run `go build ./... && go vet ./... && go test -race ./...`
4. Commit and open a Pull Request

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See [LICENSE](#) for details.